

Tide retrospective: recovery in dimension one

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

The germbij Theorem 7.3 arc established *identifiability*: the expansion data determines the loss germ. The note's Section 7.4 goes further and *recovers* the germ constructively; in dimension one, the leading exponent of the partition function recovers the degree $2k$ of an isolated minimum and the moments recover the coefficients. This tide formalises the smallest real instance of that claim, opening a constructive-recovery thread on the seabed whose closed forms (`partitionFunction_kthPotential` and the `kth_moment_even` family) were laid down in the May tides.

2 The thing formalised

Three declarations in `Laplace/OneD/Recovery.lean`, sorry-free with zero warnings. The rigidity lemma¹: if $\alpha_1 t^{\beta_1} = \alpha_2 t^{\beta_2}$ for all $t \geq T$ with $\alpha_1 > 0$, then $\beta_1 = \beta_2$ and $\alpha_1 = \alpha_2$ (positivity of α_2 is not needed). The scaling identity² $Z_{aL}(t) = Z_L(at)$. And the recovery theorem³: for potentials $a_i x^{2k_i} / (2k_i)!$ with $a_i > 0$, $k_i \geq 1$, if the partition functions agree on a ray $[T, \infty)$, $T > 0$, then $k_1 = k_2$ and $a_1 = a_2$. The exponent recovers the degree, the coefficient recovers the scale: the note's Section 7.4 statement for the zeroth moment.

3 Proof strategy

The closed form gives $Z(t) = \alpha t^{-1/(2k)}$ with $\alpha = \frac{1}{k} ((2k)!/a)^{1/(2k)} \Gamma(\frac{1}{2k})$, keeping $((2k)!/a)^{1/(2k)}$ as a single base per the consult's advice (splitting it multiplies the `rpow` bookkeeping for no benefit). For the rigidity lemma, sampling at $t_0 = \max(T, 2)$ and $t_0 \cdot t_0$ and writing $X = t_0^{\beta_1}$, $Y = t_0^{\beta_2}$ turns the two equations into $\alpha_1 X = \alpha_2 Y$ and $\alpha_1 X^2 = \alpha_2 Y^2$, whence $(\alpha_1 X)(X - Y) = 0$ by a one-line `linear_combination`; $X = Y$ forces $\beta_1 = \beta_2$ by strict monotonicity of the exponent at base $t_0 > 1$, and the coefficients cancel. No logarithms anywhere. The recovery theorem is then two injectivity steps: $k \mapsto -1/(2k)$ on positive naturals, and the base of a positive-exponent `rpow` on nonnegative reals.

4 Roadblocks and resolutions

Three build iterations. Two fixes were routine (an `omega` for a doubled natural equation; replacing a `field_simp` that mangled an `rpow` into a subtype projection by deterministic `mul_left_cancel/mul_right_cancel`)

¹`eventual_power.eq`.

²`partitionFunction.smul`.

³`kth_partitionFunction_recovery`.

The instructive one: dropping the *unused* hypothesis $0 < \alpha_2$ from the rigidity lemma strengthened the statement but left α_2, β_2 unconstrained at the application site, and the elaborator unified them with the left-hand values before the proof body could pin them, producing a vacuous goal. The repair is to pass the right-hand data explicitly $((:= \dots) \ (:= \dots))$ when the only anchor hypothesis mentions one side.

5 What was Mathlib, what was new

Mathlib: `mul_rpow`, strict monotonicity of `rpow` in the exponent, base injectivity `rpow_left_injOn`, positivity of `Gamma`. Seabed: the closed form from the May monomial tides, consumed here for the first time by a later tide. New: the sample-point elimination and the packaging.

6 Lessons

Dropping an unused hypothesis is not free when it was the elaborator's only anchor for an implicit argument: check the call sites. The two-point sampling trick is a good template for rigidity statements about power laws; it avoids both logarithms and limits.

7 Follow-ups

- The note's full one-dimensional induction: higher moments recover the subleading coefficients a_j of a general analytic potential, order by order. The seabed's `kth_moment_even` closed forms cover the base case; the inductive step needs the scaled covariance pairing.
- The semi-quasi-homogeneous weight recovery $(q_i = (\ell(2e_i) - \ell(0))/2)$ once a multivariate moment layer exists.